

# SEED STENCILS FOR DIFFUSION THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

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ABSTRACT. *[Stub, #45: the three regimes of a diffusion coefficient (smooth, a jump, an edge of width  $\delta$  below the node spacing  $h$ ); what standard stencils and the 2016 jump construction do through the sub-grid edge; the seeds and their  $\delta \rightarrow 0$  limit; the 1-D and 2-D results, elliptic and parabolic, against the coefficient treatments in use.]*

## 1. INTRODUCTION

*[Stub, #45:  $u_t = \nabla \cdot (\alpha \nabla u)$  with a coefficient that is smooth, jumps, or changes over a width  $\delta < h$ ; the knee of standard stencils at  $h \approx \delta$ ; the 2016–2017 interface-aware RBF-FD construction and its role as the  $\delta \rightarrow 0$  limit; the seeds, whose basis is continued through the edge by ODEs; what is measured, and against what.]*

**Contributions.** *[Stub, #45: the contributions, in the words LITERATURE.md section 6 allows (#43).]*

**1.1. Relation to prior work.** *[Stub, #43 supplies this paragraph (LITERATURE.md section 6a); #45 places it.]*

**Outline.** *[Stub, #45: one sentence per section.]*

## 2. SETTING

**2.1. The problem and its interface conditions.** *[Stub, #45: the heat equation and its equilibrium; continuity of  $u$  and of the flux  $\alpha \partial_n u$  across a jump; the smooth sub-grid edge of width  $\delta$  and the three regimes; the norms used.]*

**2.2. One dimension.** *[Stub, #45: equispaced grids, the naive operator assembled as  $D_x A D_x$  (dissertation eq. 76), BD4 with  $\Delta t \propto h$ , the media and test problems.]*

**2.3. Two dimensions.** *[Stub, #45: the  $x$ -periodic strip, scattered nodes with rows straddling each interface, Gaussian RBF-FD with polynomial augmentation (42 nodes and degree 5 in the bulk, 30 and degree 4 at interfaces and boundaries), Dirichlet rows, the direct and iterative solvers.]*

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**2.4. Interface-aware stencils for a jump.** *[Stub, #45: the 2016–2017 construction as background: the translated polynomial basis, the continuity matrices with curvature, the warped RBFs.]*

**2.5. Scope of the implementation.** *[Stub, #45: the Python port and the published figures it reproduces (port notes); what it leaves out (the cornered interface), stated as scope.]*

### 3. SEEDS FOR THE DIFFUSION OPERATOR IN ONE DIMENSION

**3.1. Seeds as the profiles of solutions polynomial in time.** *[Stub, #46: the chain of seeds anchored at the evaluation node, continued through the edge in first-order form so that  $\alpha$  is never differentiated; the seed-basis figure.]*

**3.2. Nested kernels and the jump as the limit.** *[Stub, #46: the nested kernels of the operator; the translated basis of Section 2.4 as the  $\delta \rightarrow 0$  limit.]*

**3.3. Well-posedness.** *[Stub, #46: the seeds form an extended complete Chebyshev system, so every stencil solve is nonsingular.]*

**3.4. Truncation error.** *[Stub, #46: third order locally on the seeded rows, fourth globally; the two readings of the bound.]*

**3.5. The elliptic problem in one dimension.** *[Stub, #46: the exactness remark and its finite-volume twin, citing the literature #43 finds.]*

### 4. ONE-DIMENSIONAL RESULTS

**4.1. Media and references.** *[Stub, #47: the smooth-edged media and the references at every  $\delta$ .]*

**4.2. The parabolic knee.** *[Stub, #47: the naive knee at  $h \approx \delta$ ; the snapshot of the four regimes on one grid.]*

**4.3. The jump construction on a smooth edge.** *[Stub, #47: the  $\delta = 0$  construction's error on a smooth edge.]*

**4.4. The seeds.** *[Stub, #47: the seeds' rates at every  $\delta$ .]*

**4.5. Coefficient treatments.** *[Stub, #47: the treatments' table and the ranking; the elliptic sanity figure as a remark.]*

### 5. SEEDS IN TWO DIMENSIONS

**5.1. A straight feature.** *[Stub, #48: the local frame and the anchor; seeds polynomial in the tangential coordinate with coefficient functions marched along the normal.]*

**5.2. Stencil assembly.** *[Stub, #48: the seeds in the RBF-FD system, the warp they bring with them, which rows are seeded; the seed-section figure.]*

**5.3. Curved and tangentially varying features.** *[Stub, #48: why a frozen normal profile is inconsistent where  $\alpha$  varies along the edge; the tangential chain in the curve's own coordinates.]*

**5.4. A thin resistive layer.** *[Stub, #48: the double edge of a ring: its exact width, series composition, the flux seeds, the warp on rows anchored inside the layer.]*

## 6. TWO-DIMENSIONAL RESULTS

**6.1. Test problems and references.** *[Stub, #49: the flat band, the sine pair, the ring; the separable, product-grid and reference-free references (stiff 4.1, 4.6, 4.8; statement 1).]*

**6.2. Naive baseline.** *[Stub, #49: the knee on scattered nodes and the jump construction on its floor (stiff 4.2; statements 2–3).]*

**6.3. Solvability and spectra.** *[Stub, #49: the seed rows in the global matrix (stiff 4.4; statement 6).]*

**6.4. The flat sweep.** *[Stub, #49: the seeds against the baselines at every  $(n, \delta)$ , the warp, the elliptic ranking (stiff 4.5; statements 4, 5, 7).]*

**6.5. The curved feature.** *[Stub, #49: the frozen profile and the tangential chain on the sine pair (stiff 4.6–4.7; statement 8).]*

**6.6. The ring and the extremizing sweep.** *[Stub, #49: the twins of EABE Fig. 19–20 and the smooth ring (stiff 4.8, 4.10; statement 9).]*

**6.7. Coefficient treatments.** *[Stub, #49: the treatments on scattered nodes (stiff 4.9; statement 10).]*

**6.8. A snapshot.** *[Stub, #49: the four regimes on one node set (stiff 5.1; statement 11).]*

## 7. DISCUSSION

**7.1. Limitations.** *[Stub, #50: corners, extreme contrasts, the cost in three dimensions, what the experiments leave open.]*

**7.2. Future work.** *[Stub, #50: in the order to do it.]*

**7.3. Conclusions.** *[Stub, #50: LITERATURE.md section 6b, verbatim.]*

## ACKNOWLEDGMENTS

*[Stub, #52: acknowledgments; where the code and data live.]*

## TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

*[Stub, #52.]*

## REFERENCES

- [Mar16] Bradley P. Martin, *Application of RBF-FD to wave and heat transport problems in domains with interfaces*, Ph.D. thesis, University of Colorado Boulder, 2016, ProQuest 10151046.
- [Mar26] ———, *Seed stencils: high-order finite differences and RBF-FD through material edges too steep for the grid*, 2026, arXiv preprint.
- [MF17a] Bradley Martin and Bengt Fornberg, *Seismic modeling with radial basis function-generated finite differences (RBF-FD) – a simplified treatment of interfaces*, Journal of Computational Physics **335** (2017), 828–845.
- [MF17b] ———, *Using radial basis function-generated finite differences (RBF-FD) to solve heat transfer equilibrium problems in domains with interfaces*, Engineering Analysis with Boundary Elements **79** (2017), 38–48.

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